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EXP 10: CRACK FTP CREDENTIALS USING A DICTIONARY ATTACK.

AIM:

To perform a dictionary-based attack against an FTP service in the authorized Metasploitable 2 lab environment using Hydra and verify the recovered credentials through a manual FTP login.

ALGORITHM:

1. Verify connectivity to the target machine at 192.168.56.101 using ping.

2. Scan TCP port 21 with Nmap to identify the FTP service and its version.

3. Create users.txt containing admin, msfadmin, root, and user.

4. Create passwords.txt containing password, msfadmin, 123456, and admin.

5. Run Hydra against the FTP service using the username and password wordlists.

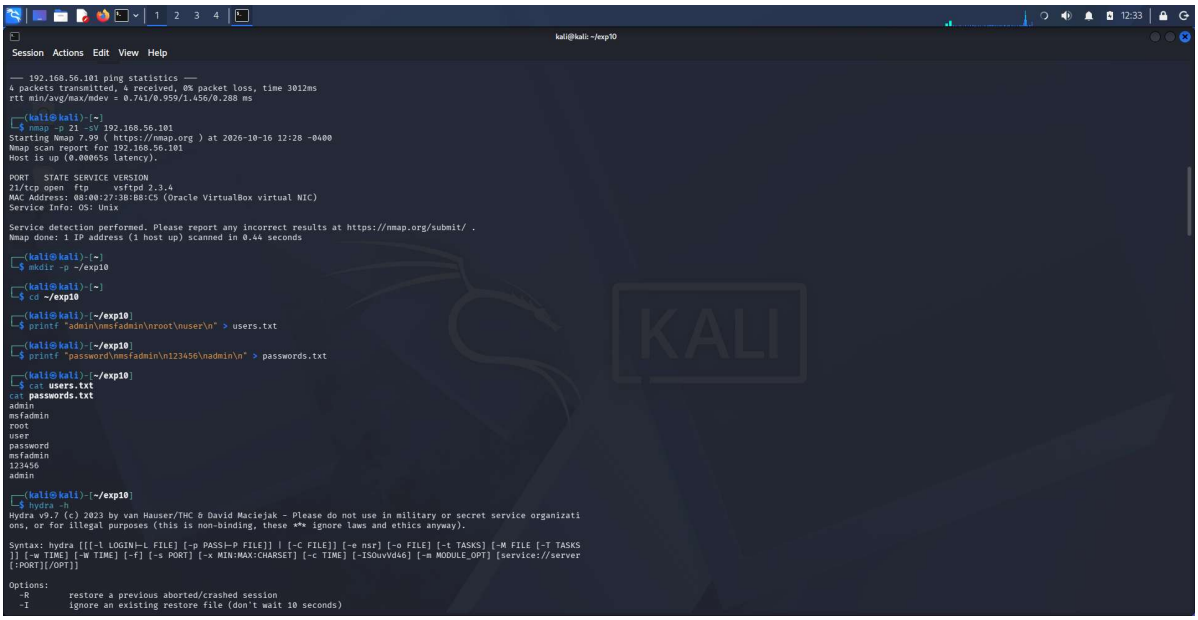
6. Record the valid credential pair reported by Hydra.

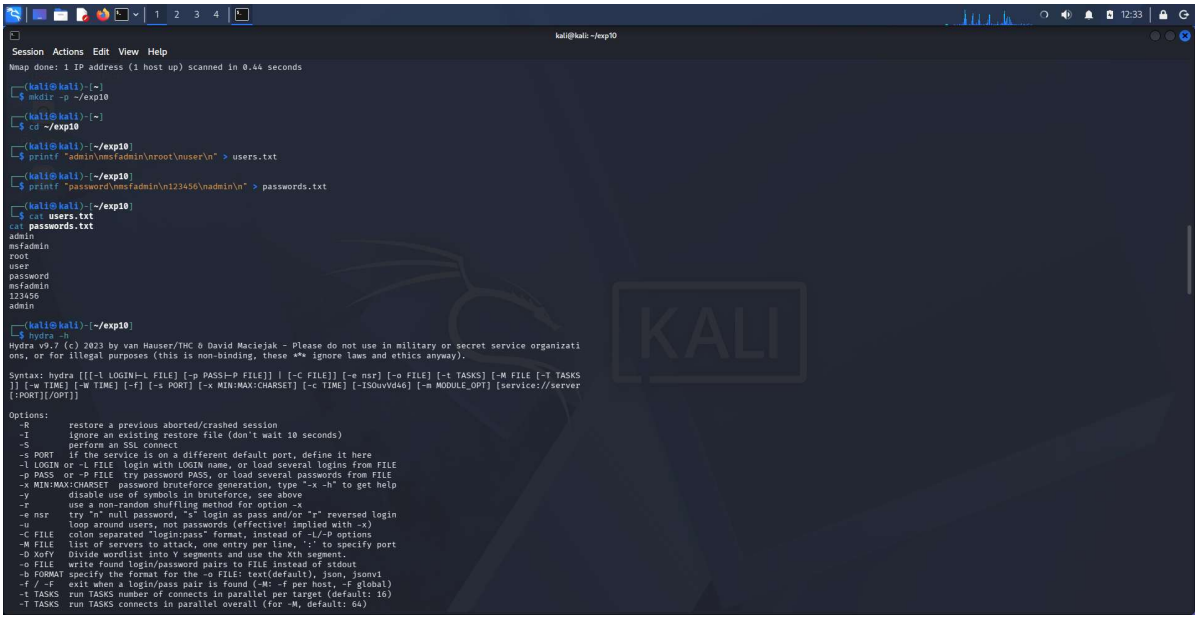
7. Connect manually with the FTP client and verify that the recovered credentials are accepted, then exit the FTP session.

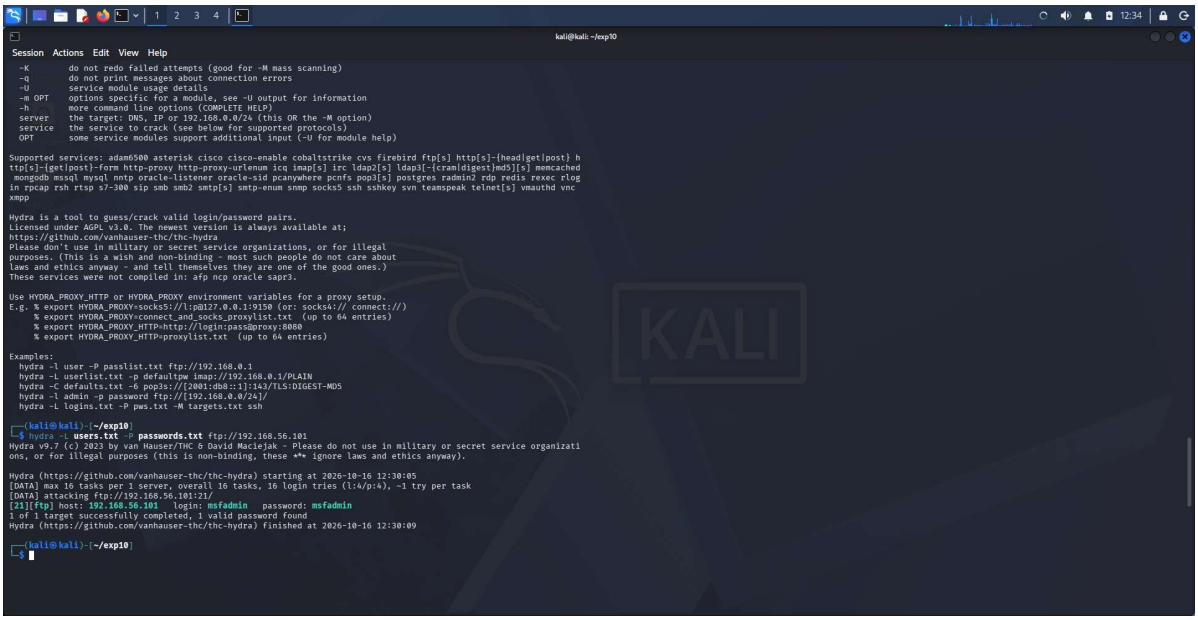
TOOLS REQUIRED:

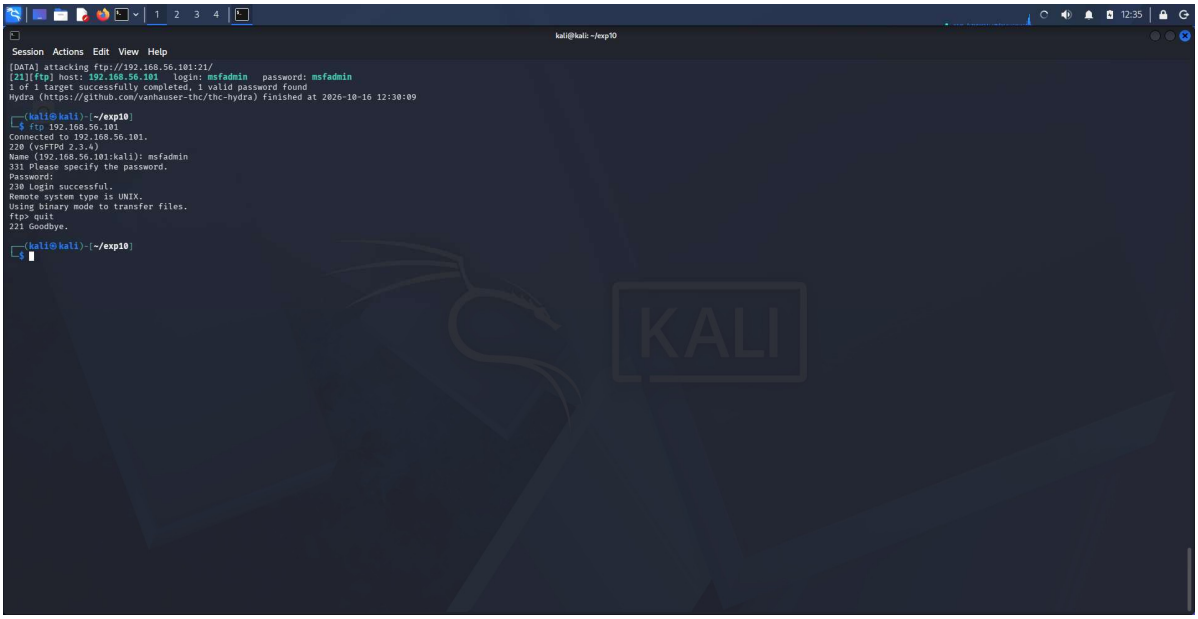
- Kali Linux
- Metasploitable 2 target machine
- Nmap
- Hydra
- Custom username and password wordlists
- FTP client

OUTPUT:


kali@kali: ~/exp10
Session Actions Edit View Help
── 192.168.56.101 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3012ms
rtt min/avg/max/mdev = 0.741/0.959/1.456/0.288 ms
┌──(kali㉿kali)-[~]
└─$ nmap -p 21 -sV 192.168.56.101
Starting Nmap 7.99 (https://nmap.org) at 2026-10-16 12:28 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00065s latency).
PORT STATE SERVICE VERSION
21/tcp open ftp vsftpd 2.3.4
MAC Address: 08:00:27:3B:B8:C5 (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.44 seconds
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/exp10
┌──(kali㉿kali)-[~]
└─$ cd ~/exp10
┌──(kali㉿kali)-[~/exp10]
└─$ printf "admin\nmsfadmin\nroot\nuser\n" > users.txt
┌──(kali㉿kali)-[~/exp10]
└─$ printf "password\nmsfadmin\n123456\nadmin\n" > passwords.txt
┌──(kali㉿kali)-[~/exp10]
└─$ cat users.txt
cat passwords.txt
admin
msfadmin
root
user
password
msfadmin
123456
admin
┌──(kali㉿kali)-[~/exp10]
└─$ hydra -h
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizati
ons, or for illegal purposes (this is non-binding, these **s ignore laws and ethics anyway).
Syntax: hydra [[[-l LOGIN|-L FILE] [-p PASS|-P FILE]] | [-C FILE]] [-e nsr] [-o FILE] [-t TASKS] [-M FILE [-T TASKS
]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX:CHARSET] [-c TIME] [-ISOuvVd46] [-m MODULE_OPT] [service://server
[:PORT][/OPT]]
Options:
-R restore a previous aborted/crashed session
-I ignore an existing restore file (don't wait 10 seconds)


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Nmap done: 1 IP address (1 host up) scanned in 0.44 seconds
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┌──(kali㉿kali)-[~/exp10]
└─$ cat users.txt
cat passwords.txt
admin
msfadmin
root
user
password
msfadmin
123456
admin
┌──(kali㉿kali)-[~/exp10]
└─$ hydra -h
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]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX:CHARSET] [-c TIME] [-ISOuvVd46] [-m MODULE_OPT] [service://server
[:PORT][/OPT]]
Options:
-R restore a previous aborted/crashed session
-I ignore an existing restore file (don't wait 10 seconds)
-S perform an SSL connect
-s PORT if the service is on a different default port, define it here
-l LOGIN or -L FILE login with LOGIN name, or load several logins from FILE
-p PASS or -P FILE try password PASS, or load several passwords from FILE
-x MIN:MAX:CHARSET password bruteforce generation, type "-x -h" to get help
-y disable use of symbols in bruteforce, see above
-r use a non-random shuffling method for option -x
-e nsr try "n" null password, "s" login as pass and/or "r" reversed login
-u loop around users, not passwords (effective! implied with -x)
-C FILE colon separated "login:pass" format, instead of -L/-P options
-M FILE list of servers to attack, one entry per line, ':' to specify port
-D XofY Divide wordlist into Y segments and use the Xth segment.
-o FILE write found login/password pairs to FILE instead of stdout
-b FORMAT specify the format for the -o FILE: text(default), json, jsonv1
-f / -F exit when a login/pass pair is found (-M: -f per host, -F global)
-t TASKS run TASKS number of connects in parallel per target (default: 16)
-T TASKS run TASKS connects in parallel overall (for -M, default: 64)





RESULT:

The FTP dictionary attack was completed successfully in the isolated lab environment. Nmap identified the FTP service running on 192.168.56.101 at port 21 as vsftpd 2.3.4. Hydra found the credential pair msfadmin / msfadmin from the supplied wordlists, and a manual FTP connection confirmed successful authentication. This demonstrates why weak or default credentials should be replaced with strong, unique passwords.